

01 Start with the side and crossing

Write a localization argument: side + level + medial/lateral or cortical/deep position + injured structures + plausible supply. Examination suggests a region; imaging establishes lesion extent. No symptom combination specifies a unique millimeter coordinate.

Four questions

1. Which functions failed? Separate weakness from ataxia, aphasia from dysarthria, sensory loss from neglect, and reduced awareness from inability to respond.
2. Which finding sets the level? A local cranial motor sign plus opposite limb findings favors brainstem; language, neglect or cortical sensory signs favor a hemispheric network.
3. Which adjacent structures explain the combination? Add crossing status to each. A single sign, such as pure motor weakness or hemianopia, has several possible locations.
4. Does the vascular field fit? Name plausible branches, then check imaging. A syndrome name does not identify the occluded artery or stroke mechanism.

Crossing ledger

Structure / segment	Main laterality rule	Important exception
Corticospinal tract	Above pyramidal crossing: opposite limbs. Below crossing: same-side limbs.	At the decussation, patterns can be mixed or bilateral.
Dorsal columns / nuclei	Same-side vibration and position until internal arcuate fibers cross in caudal medulla.	A nuclear lesion differs from a postcrossing medial lemniscal lesion.
Medial lemniscus	Opposite body after sensory decussation; ascends to VPL.	Sensory ataxia can resemble cerebellar ataxia.
Spinothalamic system	Opposite body in brainstem; fibers have crossed in spinal cord.	This is a body pathway; analyze facial sensation separately.
Spinal trigeminal system	Same-side facial pain/temperature loss at tract/nucleus.	Ascending crossed trigeminothalamic injury affects opposite face.
Cerebellar pathways	Usually same-side incoordination before superior-peduncle crossing.	Postcrossing midbrain cerebellar output: opposite ataxia.
Cranial motor structures	Many nuclear/fascicular injuries affect same-side muscles.	IV nucleus affects opposite superior oblique; III nuclei have crossed/bilateral components.

In the app: Begin in Lesion explorer. Set side before adding structures.

Supporting reading: UTHealth: [somatosensory pathways](#); UTHealth: [ocular motor control](#); UTHealth / Knierim: [cerebellum](#).

02 Medulla: distinguish the three levels

Localize relative to the pyramid, medial lemniscus, olive and dorsolateral sensory/cranial structures. Crossed face-body pain loss is a useful lateral pattern; the full textbook syndrome need not be present.

Section landmarks

Map / level	Fiber location	Localization consequence
M1: pyramidal decussation	Motor fibers cross ventrally and turn dorsolaterally. Dorsal columns remain uncrossed.	Avoid a rigid side rule for weakness when injury centers on the crossing.
M2: sensory decussation	Gracile nucleus is dorsomedial; cuneate is lateral. Internal arcuate fibers cross and form medial lemniscus.	Dorsal nuclear loss is ipsilateral; the newly formed lemniscus carries contralateral information.
M3: open medulla / olive	Pyramids are ventral; medial lemniscus is dorsal to them and medial to olive. XII fascicles travel ventrally.	XII lower motor signs plus opposite weakness and position loss support medial medulla.

At the olive: explain each sign

Injured structure	Possible manifestation	Anatomical neighborhood
Pyramid + medial lemniscus	Contralateral weakness + vibration/position loss.	Ventral / paramedian medulla.
XII nucleus / fascicles	Ipsilateral tongue weakness; tongue points toward the weak side on protrusion.	Nucleus by fourth-ventricular floor; fascicles descend ventrally.
Nucleus ambiguus	Ipsilateral palatal/vocal-fold weakness, dysphagia, hoarseness.	Deep lateral reticular formation, dorsal to olive.
Spinal V + spinothalamic	Ipsilateral facial and contralateral body pain/temperature loss.	Lateral medulla; face and body pathways differ in crossing status.
Inferior peduncle / vestibular / sympathetic fibers	Ipsilateral ataxia or lateropulsion, vestibular signs, ipsilateral Horner syndrome.	Dorsolateral and lateral medulla; combinations depend on extent.

Supply

Medial medulla: vertebral paramedian and anterior spinal contributions. Lateral medulla: vertebral and/or PICA medullary branches. Dorsal closed medulla: posterior spinal/vertebral branches. Wallenberg is not synonymous with proven PICA occlusion.

In the app: Compare M1, M2 and M3; load medial or lateral medullary examples in M3.

Supporting reading: [UTHealth: neuroanatomy sections and major landmarks](#); [UTHealth: somatosensory pathways](#); [UBC brain micrographs](#).

03 Pons: separate gaze, facial and long tracts

The basis pontis contains descending corticofugal bundles interspersed with pontine nuclei and transverse fibers. The tegmentum contains cranial and ascending systems. Pure motor pontine infarcts may lack a cranial nerve sign.

Section landmarks

Map / level	Structures to find	Useful distinction
P1: facial colliculus	VI nucleus near dorsal midline; VII fibers loop around it. MLF is paramedian; facial motor nucleus lies farther ventrolaterally.	Facial-colliculus injury involves VII fascicles, not necessarily the VII nucleus.
P2: trigeminal level	Motor V is medial to principal sensory V. MCP is lateral. Medial lemniscus is a transverse band dorsal to basis.	Separate ipsilateral trigeminal nuclear signs from crossed ascending trigeminal deficits.
P3: superior peduncles	SCPs converge dorsolaterally; locus coeruleus lies near lateral ventricular floor; arousal pathways traverse tegmentum.	SCP injury below its midbrain crossing usually causes ipsilateral ataxia.

Horizontal gaze: four different lesions

Left-sided injury	What fails	What distinguishes it
VI fascicle / nerve	Left eye abduction.	The right eye can still adduct on attempted left gaze.
VI nucleus	Conjugate gaze left: left abduction and right adduction.	Nuclear motor neurons and interneurons are involved.
PPRF	Saccades toward the left.	Reflex eye movements can differ from those in a VI nuclear lesion.
MLF	Left adduction during right gaze; right abducting nystagmus may occur.	Internuclear ophthalmoplegia. Convergence can be preserved; it is not invariably so.
VI nucleus/PPRF + MLF	Neither eye looks left; on right gaze only the right eye abducts.	One-and-a-half. Add left peripheral-pattern facial weakness for eight-and-a-half.

Supply and larger patterns

Paramedian pons: basilar perforators. Lateral caudal pons/MCP: AICA and basilar circumferential branches, with variable inner-ear supply. Rostral lateral pons/SCP: SCA-related branches. Extensive bilateral ventral CST and corticobulbar injury with preserved awareness can produce a locked-in pattern.

In the app: In P1, compare VI nucleus with VI fascicle, then load one-and-a-half or eight-and-a-half.

Supporting reading: UTHealth: [ocular motor control](#); Clinical spectrum of pontine infarction; Lee et al., 2002: [deafness and AICA infarction](#).

04 Midbrain: tectum, tegmentum and crus

Dorsal tectum lies behind the aqueduct. Tegmentum lies between the aqueduct and substantia nigra; the crus lies ventral to the nigra. The same III sign can accompany very different neighboring tract injuries.

Section landmarks

Map / level	Important fibers and nuclei	Crossing / deficit clue
B1: inferior colliculus	IV nucleus ventral to aqueduct; SCP decussation farther ventral in central tegmentum; lateral lemniscus enters inferior colliculus.	IV axons cross before dorsal exit. Central SCP crossing injury can cause bilateral ataxia.
B2: superior colliculus	III nuclei near midline; III fascicles travel through tegmentum. Red nucleus is dorsal to nigra; medial lemniscus lies laterally.	III fascicle injury generally produces ipsilateral ocular deficits; crossed cerebellar output gives contralateral ataxia.
B3: pretectal junction	Pretectum, posterior commissure, rostral interstitial MLF and interstitial nucleus of Cajal; adjacent arousal networks.	Vertical saccade and gaze-holding deficits implicate a rostral network, not simply the III nerve.

Nuclear and fascicular exceptions

Structure	Possible manifestation	Do not confuse with
III fascicles	Ipsilateral ptosis, impaired adduction/elevation/depression; variable pupil involvement.	An isolated nuclear pattern or a peripheral III lesion cannot be inferred from ptosis alone.
III nuclear complex	Contralateral superior rectus weakness; bilateral ptosis can coexist with other ocular deficits.	Each nucleus does not simply control only its ipsilateral eye.
IV nucleus	Contralateral superior oblique weakness with vertical/torsional diplopia.	The emerged IV nerve supplies the ipsilateral superior oblique.
Dorsal midbrain network	Upgaze impairment, light-near dissociation or convergence-retraction movements may occur.	The syndrome has nonvascular causes as well as stroke.

Think in components, then use the eponym

Weber: III fascicles + crus → ipsilateral III palsy and contralateral weakness. Claude: III fascicles + tegmental cerebellar outflow → ipsilateral III palsy and contralateral ataxia. Tremor or involuntary movements in a broader tegmental lesion may be described as Benedikt-type. Boundaries overlap.

Supply: PCA peduncular, paramedian, circumferential and collicular branches, with basilar-tip, SCA and communicating-artery contributions varying by site.

In the app: Compare Weber in B2 with Claude; compare III fascicles with the III nuclear complex.

Supporting reading: UTHHealth: [ocular motor control](#); UTHHealth / Knierim: [cerebellum](#); 3T whole-brain microscopy: [brain stem](#).

05 Thalamus and capsule: compact relays and fibers

Thalamic vascular fields do not coincide precisely with nuclear borders. Pure sensory stroke is not uniquely thalamic, and pure motor stroke is not uniquely capsular. A lesion may interrupt passing fibers or connected networks.

Thalamic relays and vascular fields

Region / nuclei	Possible loss of function	Typical supply / caveat
Anterior nuclei + mammillothalamic tract	New-learning and orientation problems; dominant involvement can emphasize verbal memory.	Tuberothalamic/polar from PCom; variant paramedian contribution.
Mediodorsal + intralaminar/midline	Executive/memory changes; bilateral paramedian injury can markedly reduce alertness.	PCA P1 paramedian branches. Percheron is one possible shared-perforator variant.
VPL / VPM	Opposite body / face sensory loss. VPM lies medial to VPL.	Inferolateral/thalamogeniculate branches, principally PCA P2; overlap exists.
VA / VL	Motor planning, initiation or movement/coordination disturbances.	Tuberothalamic, paramedian and inferolateral contributions vary by nucleus.
Pulvinar / geniculate region	Attention/visual-integration abnormalities; LGN injury can cause opposite homonymous field loss.	Posterior choroidal branches; LGN also receives anterior choroidal supply.

Capsular geometry

Part	What runs there / neighbors	Lesion clue
Anterior limb	Between caudate head and lentiform nucleus; frontal projection systems.	Abulia/executive change; Heubner and lenticulostriate contributions.
Genu	Bend of capsule; corticobulbar fibers in genu and adjacent regions.	Opposite lower facial/tongue weakness and dysarthria may occur.
Posterior limb	Between thalamus and lentiform nucleus; compact motor and sensory projections.	Opposite motor and/or sensory loss. Lenticulostriate and anterior choroidal supply overlaps.
Retrolenticular / sublenticular	Behind lentiform: optic projections. Beneath lentiform: auditory projections, with nearby visual fibers.	Field loss or auditory processing change; these projected compartments are not rigid isolated bundles.

In the app: Use T1 for anterior/medial relays, T2 for ventral/posterior relays, D1 for capsule.

Supporting reading: Schmahmann, 2003: vascular syndromes of the thalamus; Isolated thalamic stroke, 2021 cohort; UTHHealth: internal capsule; UBC: blood supply of the brain.

06 Cortex: name the gyrus and the network

Functional boundaries vary. Brodmann numbers are useful orientation labels, not exact stroke segmentation. Set language dominance explicitly; handedness alone does not establish it.

Regional checklist

Anatomical region	Possible manifestation	Vascular context
Lateral precentral gyrus, BA4	Opposite face/arm weakness; small hand-knob lesions may mimic a peripheral palsy.	MCA superior division / central branches.
Postcentral gyrus, BA3/1/2	Opposite primary or discriminative sensory loss; assess cortical sensory tasks only when primary sensation and comprehension allow.	MCA parietal branches.
Paracentral lobule	Opposite leg-predominant motor and/or sensory loss.	ACA pericallosal/callosomarginal branches.
Supplementary motor / cingulate	Reduced initiation, akinesia or abulia; dominant injury may reduce spontaneous speech.	ACA fields; severity depends strongly on bilateral extent.
Inferior frontal language network, BA44/45 region	Dominant injury may cause nonfluent language. Persistent classic Broca aphasia often needs a wider lesion.	MCA superior division.
Posterior temporal language network	Dominant injury may cause fluent paraphasic speech with impaired comprehension.	MCA inferior division; language is distributed beyond one gyrus.
Angular / supramarginal gyri, BA39/40	Dominant symbolic, writing/calculation, phonological or praxis deficits; full Gerstmann tetrad is not obligatory.	MCA inferior/parietal branches; distal PCA overlap.
Temporoparietal attention network	Right-sided injury can cause left neglect/extinction or anosognosia. Neglect and hemianopia can coexist.	MCA posterior/inferior fields; deep networks also contribute.
Frontal eye field	Acute destructive injury may bias gaze toward the injured hemisphere.	MCA superior division; seizure-related deviation can differ.
Calcarine / medial temporal / ventral occipitotemporal	Field, learning, reading or recognition deficits, depending on structures and side.	PCA surface branches, with regional anterior choroidal/MCA overlap.

Avoid a category error

Aphasia concerns language; dysarthria concerns speech execution. Neglect concerns attention; hemianopia concerns visual fields. Apraxia is impaired learned action not explained by basic weakness, ataxia or poor comprehension.

In the app: C1 is the lateral surface; C2 is medial/inferior. Deep fibers on these maps are projections.

Supporting reading: UBC Functional Neuroanatomy; UTHHealth: ocular motor control; UBC: blood supply of the brain; Brandt et al., 2000: PCA territory infarcts.

07 Visual, auditory and cerebellar localization

A visual field defect narrows a pathway but often does not identify a unique level. Cerebellar signs must be separated from weakness and sensory ataxia. Unilateral central auditory injury usually differs from peripheral hearing loss.

Visual field comparison

Left-sided pathway injury	Expected field pattern	What helps distinguish it
Optic tract / LGN	Right homonymous defect; LGN can give sectoral loss.	Pupillary and associated neurological findings can help; field shape alone is insufficient.
Meyer loop, temporal white matter	Right superior homonymous quadrantanopia.	Temporal language/memory or adjacent cortical findings may support this site.
Parietal optic radiation	Right inferior homonymous quadrantanopia.	Parietal sensory/attention findings may accompany it.
Upper calcarine bank / cuneus	Right inferior homonymous quadrantanopia if selective.	Same quadrant as a parietal radiation lesion; imaging separates them.
Lower calcarine bank / lingual region	Right superior homonymous quadrantanopia if selective.	Same quadrant as Meyer loop; a larger calcarine lesion can give hemianopia.
Dominant occipital + splenial disconnection	Alexia without agraphia may accompany field loss.	Reading failure is not established by occipital damage alone; test writing and language.

Cerebellar pathways

Structure	Route / fiber position	Clinical implication
Inferior peduncle	Dorsolateral medulla; major spinal/olivary/vestibular inputs.	Usually ipsilateral incoordination and imbalance.
Middle peduncle	Lateral pons; crossed pontocerebellar input.	Lesion at the peduncle usually produces ipsilateral ataxia.
Superior peduncle	Dorsolateral rostral pons → caudal midbrain crossing → thalamic/red-nuclear pathways.	Ipsilateral before crossing; contralateral after crossing; crossing-centered injury can be bilateral.
Vermis / hemisphere / vestibulocerebellum	Midline posture; limb coordination; vestibular and ocular motor functions, respectively.	Truncal versus limb versus eye-movement emphasis helps define the involved network.

Hearing clue

AICA-related vertigo with new hearing loss may reflect labyrinthine ischemia. A brainstem entry-zone lesion can affect same-side hearing, whereas more rostral auditory pathways carry bilateral information. Bilateral central injury can cause severe central hearing dysfunction.

In the app: Compare Meyer loop, parietal radiation, cuneus and lingual cortex; then compare SCP with its crossing.

Supporting reading: University of Michigan: visual pathway lesions; UTHealth / Knierim: cerebellum; Lee et al., 2002: deafness and AICA infarction.

08 Advanced fibers: compact route notes

These systems are important anatomically, but many lack a dependable isolated human stroke syndrome. Their blood supply changes along the course; do not assign a single artery to an entire long pathway.

Motor and cerebellar fibers

Pathway	Course and crossing	Why it matters
Dorsal spinocerebellar	Clarke relay → ipsilateral lateral cord → inferior peduncle.	Lower-body unconscious proprioceptive input; differs from conscious DCML sensation.
Cuneocerebellar	Accessory cuneate relay → ipsilateral inferior peduncle.	Upper-limb input; accessory cuneate is distinct from the main cuneate relay.
Ventral / rostral spinocerebellar	Ventral fibers cross in cord and often recross via cerebellar circuits; rostral pathway is largely uncrossed.	Internal spinal feedback. Superior and inferior peduncular routes differ by component.
Corticopontine	Approximate medial frontal and lateral posterior-cortical crus sectors → pons → crossed pontocerebellar fibers.	Disconnection may add ataxia to weakness; crus fibers overlap.
Central tegmental tract	Red-nuclear/tegmental connections descend toward ipsilateral inferior olive.	Interruption can cause delayed palatal/oculopalatal tremor with olivary degeneration.
Rubrospinal / tectospinal	Ventral / dorsal tegmental decussations, respectively; descend toward spinal motor/orienting circuits.	Do not equate isolated injury with a dense corticospinal hemiplegia.
Reticulo- / vestibulospinal	Pontomedullary reticular/vestibular origins → overlapping bilateral and ipsilateral descending systems.	Posture, tone and lateropulsion; limited specificity for an isolated tract lesion.
Pallidothalamic	GPI → ansa lenticularis/lenticular fasciculus → Forel fields → VA/VL.	Small subthalamic fiber fields beside capsule/STN; movement effects vary.

Association and limbic fibers

Arcuate/SLF: dorsal frontotemporoparietal communication; dominant injury can impair repetition/phonology. IFOF/ILF: ventral association routes involved in semantic and visual recognition systems; they are distinct bundles, despite being grouped on the teaching map.

Uncinate links anterior temporal and ventral frontal regions. Cingulum runs in cingulate white matter. Fornix carries hippocampal projections toward diencephalon; mammillothalamic fibers connect mammillary bodies with anterior thalamus. Memory and executive effects depend on network extent.

Dorsal longitudinal fasciculus is an autonomic pathway and is not the ocular-motor MLF. Human gustatory laterality does not follow one universally accepted simple crossing rule.

In the app: Open Pathways & fibers → Advanced fibers & loops for location and vascular context along each route.

Supporting reading: [UTHealth / Knierim: cerebellum](#); [UTHealth: rubrospinal pathway](#); [UTHealth: neuroanatomy sections and major landmarks](#); [UTHealth: internal capsule](#).

09 Vascular patterns: artery to clinical ensemble

Territories overlap and collateral flow varies. Infarct location does not by itself establish embolism, small-vessel disease or hypoperfusion. Use these as anatomical hypotheses, not occlusion diagnoses.

Supratentorial and posterior-fossa map

Vessel / field	Structures at risk	Possible combination
ACA surface branches	Medial frontal, paracentral and cingulate regions.	Opposite leg-predominant deficit, reduced initiation; bilateral injury can be profound.
MCA superior division	Lateral frontal/central cortex and related fibers.	Opposite face-arm weakness; dominant nonfluent language or gaze preference may accompany.
MCA inferior division	Posterior temporal/inferior parietal networks and nearby optic fibers.	Dominant fluent aphasia or nondominant neglect; field loss may occur.
MCA lenticulostriates / ACA Heubner	Striatocapsular regions / caudate head and anterior capsule.	Compact motor deficit or executive/initiative disturbance depending on field.
Anterior choroidal	Posterior capsule and optic structures; pallidal/temporal contributions vary.	Opposite weakness, sensory loss and field loss can combine; complete triad is not required.
PCA cortical branches	Calcarine, medial temporal and posterior association/splenic regions.	Opposite homonymous field loss with variable memory, recognition or reading disturbance.
PCA / PCom perforators	Midbrain and thalamic fields.	Ocular-motor, sensory, movement, memory or arousal changes according to territory.
Vertebral / ASA / PICA medullary	Paramedian versus lateral/dorsal medulla.	Medial tongue-long-tract pattern versus lateral crossed-pain/bulbar/ataxic pattern.
Basilar perforators / AICA	Paramedian pons versus lateral caudal pons, MCP and inner-ear supply variably.	Motor/gaze pattern versus facial, vestibular, hearing and ataxic ensemble.
SCA / AICA / PICA cerebellar	Superior / anteroinferior / posteroinferior fields, with variation.	Limb/truncal/vestibular ataxia. A cerebellar infarct need not include brainstem signs.
Border zones	Distal ACA-MCA, MCA-PCA or internal deep junctional fields.	Pattern depends on distribution; border-zone location does not prove systemic hypotension.

In the app: Vascular territories provides 26 examples, each linked to the affected structures and simulator.

Supporting reading: UBC: blood supply of the brain; Schmahmann, 2003: vascular syndromes of the thalamus; Clinical spectrum of pontine infarction; Brandt et al., 2000: PCA territory infarcts; Lee et al., 2002: deafness and AICA infarction.

10 Worked localization and final self-check

For every case, name the structures before naming the syndrome. The examples below are coherent educational hypotheses; real patients may have incomplete, extended, multifocal or nonvascular patterns.

Six short cases

Findings	Best-fitting localization	Structures / plausible supply
Left facial pain loss, right body pain loss, hoarseness, left ataxia.	Left lateral medulla.	Spinal V + spinothalamic + ambiguous + inferior peduncle; vertebral/PICA branches.
Right tongue lower motor weakness; left limb weakness and position loss.	Right medial medulla.	XII fascicles + pyramid + medial lemniscus; vertebral paramedian/ASA contributions.
Neither eye looks left; only right eye abducts on right gaze; left whole-face weakness.	Left dorsal caudal pons; eight-and-a-half pattern.	VI nucleus/PPRF + left MLF + VII fascicles; basilar tegmental perforator field.
Left III palsy with right limb ataxia, little weakness.	Left midbrain tegmentum.	III fascicles + postcrossing cerebellar output; PCA tegmental branches.
Severe drowsiness, amnesia and vertical gaze impairment.	Consider bilateral paramedian thalamic/rostral midbrain network injury.	Paramedian perforators; Percheron is one variant, not a diagnosis from this triad.
Right superior homonymous quadrantanopia.	Left Meyer loop OR left lower calcarine region, among alternatives.	Temporal versus occipital accompanying signs may help; imaging resolves the level.

Before accepting your localization

Can one anatomically plausible lesion explain the positive findings? Which finding is most level-specific? Have you stated the crossing correctly?

Are apparent spared functions actually examined? Does the alternative require fewer assumptions? What imaging observation would distinguish the possibilities?

Could the lesion involve passing fibers or disconnect a network? Does a vascular pattern fit without assuming that every branch supplies an identical territory in every person?

Using the companion atlas

Open [Neurovascular-Atlas.html](#) in a browser. Choose a level, set side/dominance, inspect a marker and add structures. The 129-page [Neurovascular-Atlas.pdf](#) contains the full directories, 15 locator plates and bibliography. Keep both files beside these notes.

Scope: major clinically relevant structures and selected advanced pathways. The diagrams are approximate educational maps, not every microscopic fiber or a validated patient-specific localization system.

In the app: Study prompt: state side → level → structures → crossings → supply → strongest alternative.

Supporting reading: [UTHealth: ocular motor control](#); [Schmahmann, 2003: vascular syndromes of the thalamus](#); [University of Michigan: visual pathway lesions](#); [UTHealth: neuroanatomy sections and major landmarks](#).